ONLINE ASSESSMENT · 24 OCT 2026

Flipkart

Campus: IIT Delhi · Role: AI Engineer

Pattern: 3 coding questions · 90 minutes

1. Questions

Q1. ASCII Digit Checksum

Flipkart's messaging service tags every outgoing message with a small per-character checksum so corrupted characters can be spotted.

- You are given a message `msg` (a `char[]` in C++, a list of characters in Python).
- For each character, take its ASCII code and add up the decimal digits of that code (e.g. `'a'` → 97 → 9 + 7 = 16).

Return: print the digit sum for each character, one by one, in order.

EXAMPLES

```
Input:  msg = "Hi"
Output: 9
        6
Why:    'H' = 72 → 7+2 = 9,  'i' = 105 → 1+0+5 = 6
```

```
Input:  msg = "a!"
Output: 16
        6
Why:    'a' = 97 → 9+7 = 16,  '!' = 33 → 3+3 = 6
```

```
Input:  msg = "A"
Output: 11
Why:    single character: 'A' = 65 → 6+5 = 11
```

CONSTRAINTS

> Constraints: not recorded

Q2. Balanced-Parity Category Nodes

Flipkart's product catalogue is stored as a binary category tree. Each category has an id and a value (say, number of listed products), and an audit flags categories whose two sub-catalogues disagree in parity.

- You are given the id of the **root** node and an integer n.
- Then n lines follow, each `path node`: `path` is a string over `L` / `R` describing the route from the root (e.g. `LRLR`), and `node` is the id of the node at that position.
- Node ids are distinct but need not be consecutive (e.g. 11, 6, 22).
- You are also given an array `value`, where `value[i]` is the value of the i-th node (root first, then the nodes in input order).
- A node u is **flagged** if it has **both** a left and a right child, and the sum of values in its left child's subtree and the sum of values in its right child's subtree have **different parity** (one odd, one even).

Return: the sum of the ids of all flagged nodes.

EXAMPLES

```
Input:  root = 11, n = 4
        L 6
        R 22
        LL 4
        LR 9
        value = [3, 1, 2, 4, 6]     (nodes 11, 6, 22, 4, 9)
Output: 11
Why:    node 11: left {6,4,9} = 11 (odd), right {22} = 2 (even) → flagged.
        node 6: left {4} = 4, right {9} = 6, both even → not flagged.
```

```
Input:  root = 10, n = 6
        L 3
        R 25
        LL 7
        LR 14
        RL 2
        RR 30
        value = [1, 2, 3, 4, 5, 6, 7]     (nodes 10, 3, 25, 7, 14, 2, 30)
Output: 38
Why:    node 10: 11 vs 16 → flagged; node 3: 4 vs 5 → flagged;
        node 25: 6 vs 7 → flagged; 10 + 3 + 25 = 38
```

```
Input:  root = 5, n = 2
        L 8
        LL 1
        value = [5, 1, 2]
Output: 0
Why:    no node has two children
```

CONSTRAINTS

- $n \leq 1000$ (recalled as 100 or 1000)

Q3. Service Engineer

Mona is an on-site service engineer for a laptop manufacturer. She regularly travels across her territory visiting the sites where laptop servicing needs to be done. She works 8 hours a day, spending time either travelling to a site or providing service at the site.

You are given information about Mona's territory in the form of an $R \times C$ dimensional array $matrix$. Each cell in the $matrix$ contains a number denoting the time needed by Mona to complete the servicing at that site. In case a cell has 0, it means that no servicing is required there.

Mona can move only horizontally or vertically at a time and she takes 2 hours to move to the next cell.

Mona starts at cell $(0, 0)$ and can finish in any cell. Your job is to find the minimum number of days in which she can complete all the tasks.

INPUT FORMAT

The first line contains two integers R and C - denoting the number of rows and columns in $matrix$.

The next R lines contain C characters each - denoting the cells of $matrix$.

OUTPUT FORMAT

Print the minimum number of days in which Mona can complete all the tasks.

EXAMPLES

```
Input:  3 3
        2 0 0
        0 0 0
        0 0 1
Output: 2
Why:    servicing 2 + 1 = 3 h; route (0,0)→(2,2) = 4 units = 8 h; 11 h → 2 days
```

```
Input:  3 4
        3 0 0 0
        3 0 0 0
        0 0 5 0
Output: 3
Why:    servicing 11 h; route (0,0)→(1,0)→(2,2) = 4 units = 8 h; 19 h → 3 days
```

```
Input:  2 3
        0 0 0
        0 0 0
Output: 0
Why:    no servicing required
```

CONSTRAINTS

- $1 \leq R \leq 30$
- $1 \leq C \leq 30$
- $0 \leq matrix[i][j] \leq 1000$
- It's guaranteed the number of non-zero cells in $matrix \leq 15$.
- Time limit 1 s · Memory 256 MB

> Same as Goldman Sachs Coding Test II – "Service Engineer", except the start is fixed at (0, 0) (there, Mona can start and finish in any cell).